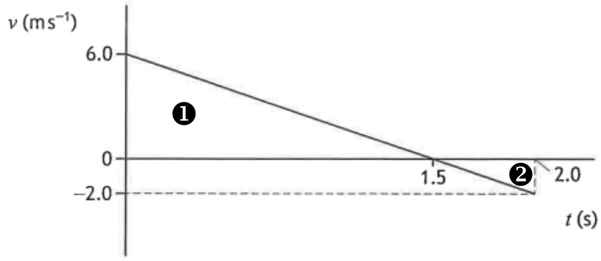
JABstem

Past Papers Nat 5 Physics 2025 Marking Scheme

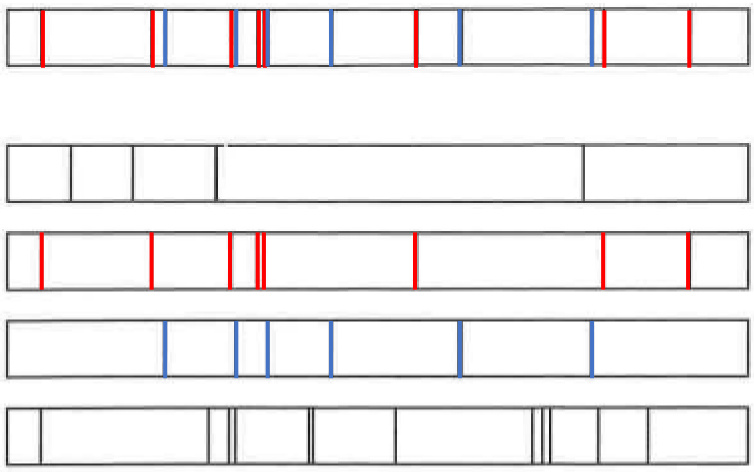
Grade Awarded	Mark Required		% candidates achieving grade
	/125	%	
A	84+	67%+	32.8%
B	70+	56%+	20.7%
C	56+	45%+	19.1%
D	42+	34%+	15.3%
No award	<42	<34%	12.1%

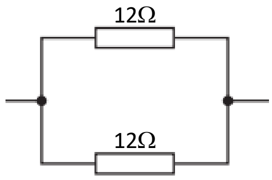
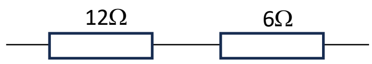
Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	15.6 /25	38.1 /75	17.3 /25

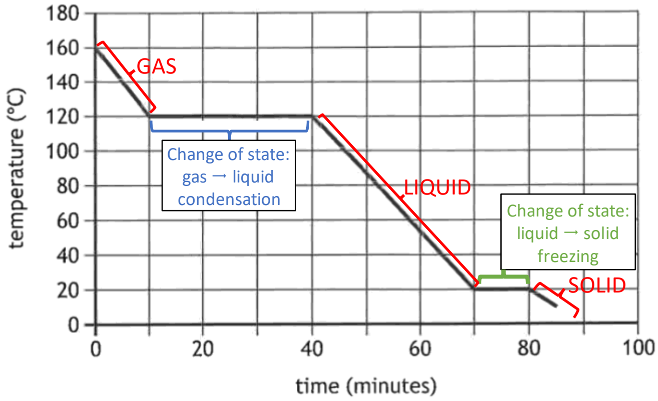
2025 Nat5 Physics Marking Scheme

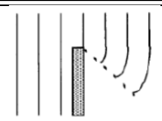
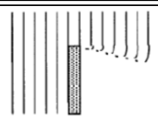
<table>
<tr><th>Question</th><th>Answer</th><th>% Correct</th><th>Physics Covered</th></tr>
<tr><td>1</td><td>B</td><td>74</td><td>
<table>
<tr><td>Scalar Quantity</td><td>energy</td><td>speed</td><td>distance</td><td>time</td><td>mass</td></tr>
<tr><td>Vector Quantity</td><td>force</td><td>velocity</td><td>displacement</td><td>acceleration</td><td>weight</td></tr>
</table>
</td></tr>
<tr><td>2</td><td>A</td><td>42</td><td>


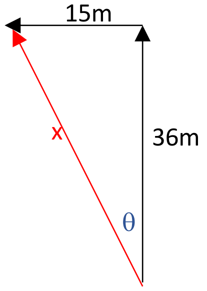
<table>
<tr><td>Area ❶</td><td>Area ❷</td><td rowspan="4">Total displacement = 4.5m – 0.5m = 4.0m</td></tr>
<tr><td>Distance = area under graph</td><td>Distance = area under graph</td></tr>
<tr><td>= $\frac{1}{2}$ x 1.5 x 6.0</td><td>= $\frac{1}{2}$ x 0.5 x -2</td></tr>
<tr><td>= 4.5m</td><td>= -0.5m</td></tr>
</table>
</td></tr>
<tr><td>3</td><td>D</td><td>59</td><td>
$a = -2\ m\ s^{-2}$ $v = 5\ m\ s^{-1}$ u =? t = 8.0 s
$a = \frac{v - u}{t}$
$-2 = \frac{5 - u}{8}$
$-16 = 5 - u$
$u = 5 + 16$
$u = 21\ m\ s^{-1}$
</td></tr>
<tr><td>4</td><td>A</td><td>65</td><td>
☑A Force of the table on the cup is balanced by the force of cup on the table
☒B Force of the cup on the Earth is balanced by the force of Earth on the cup
☒C Force of the Earth on cup is balanced by force of the cup on Earth
☒D Force of the cup on the table is balanced by force of the table un the cup
☒E Force of the table on the Earth is balanced by the force of Earth on the table
</td></tr>
<tr><td>5</td><td>E</td><td>63</td><td>
W = ? m = 180kg $g = 9.8\ N\ kg^{-1}$
$W = m\ g$
$W = 180 \times 9.8$
$F = 1764\ N$
E_W = ? F = 1764 N d = 3.0m
$E_W = F\ d$
$E_W = 1764 \times 3.0$
$F = 5292\ J$
</td></tr>
<tr><td>6</td><td>D</td><td>57</td><td>
E_k = ? m = 25kg $v = 9200\ m\ s^{-1}$
$E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 25 \times (9200)^2 = 1.058 \times 10^9\ J = 1.1 \times 10^9\ J$
</td></tr>
<tr><td>7</td><td>C</td><td>62</td><td>
v = ? $r = 6.4 \times 10^6 + 3.6 \times 10^7 = 4.24 \times 10^7\ m$ $g = 0.22\ N\ kg^{-1}$
$v = \sqrt{rg} = \sqrt{4.24 \times 10^7 \times 0.22} = \sqrt{9.33 \times 10^6} = 3054\ m\ s^{-1} = 3100\ m\ s^{-1}$
</td></tr>
</table>

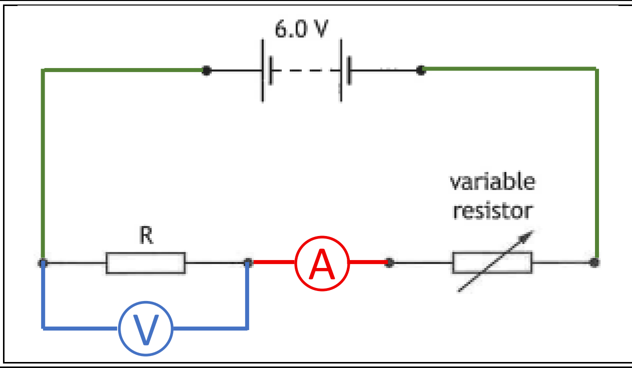
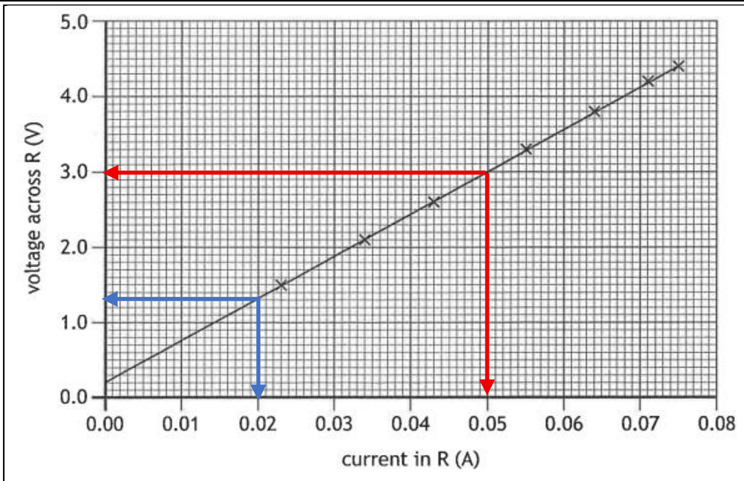
<table>
<tr><td>8</td><td>C</td><td>76</td><td>W = ?   m = 3.5kg   g = 7.7 N kg^{-1}
$W = m \times g$
$W = 3.5 \times 7.7$
W = 27 N</td></tr>
<tr><td>9</td><td>E</td><td>62</td><td>$d = v \times t$
$d = 3.0\times10^{8} \times 8.6 \times 365.25 \times 24 \times 60 \times 60$
d = 8.1×10^{16} m</td></tr>
<tr><td>10</td><td>C</td><td>91</td><td>
star
hydrogen
helium
carbon
nitrogen</td></tr>
<tr><td>11</td><td>E</td><td>87</td><td><table>
<tr><th>Trace X</th><th>Trace Y</th><th>Trace Z</th></tr>
<tr><td>d.c. signal</td><td>d.c. signal</td><td>a.c. signal</td></tr>
<tr><td>• negative charges (electrons) flow in one direction only.
• gives a constant trace on oscilloscope</td><td>• negative charges (electrons) flow in opposite direction to X
• gives a constant trace on oscilloscope</td><td>• the direction of electrons in current changes back and forth at regular intervals
• the size of the current varies with time and is not constant</td></tr>
</table></td></tr>
<tr><td>12</td><td>A</td><td>67</td><td>☑A Q+R must have same charge but opposite charge to particle
☒B Q+R must have same charge as particle is attracted to both Q+R
☒C Q+R must have same charge as particle is attracted to both Q+R
☒D Q+R must have same charge as particle is attracted to both Q+R
☒E particle must have opposite charge to Q+R for particle to be attracted to Q+R</td></tr>
<tr><td>13</td><td>D</td><td>50</td><td>Voltage in 240Ω resistor = 3.0V ∴ Voltage in unknown resistor = 12.0V - 3.0V = 9.0V
V_1 = 3.0V V_2 = 9.0V R_1 = 240V R_2 = ?
$\frac{V_1}{V_2} = \frac{R_1}{R_2}$
$\frac{3.0}{9.0} = \frac{240}{R_2}$
$R_2 \times 3.0 = 240 \times 9.0$
$R_2 = \frac{240 \times 9.0}{3.0}$
$R_2 = 720\ \Omega$</td></tr>
</table>

<table>
<tr>
<td>14</td><td>D</td><td>65</td>
<td>
<table>
<tr>
<td>Combining Parallel Resistors:

$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ (1 mark)
$\frac{1}{R_T} = \frac{1}{12} + \frac{1}{12}$ (1 mark)
$\frac{1}{R_T} = \frac{2}{12}$
$R_T = {}^{12}/_{2} = 6\ \Omega$</td>
<td>Combining Series Resistors:

$R_T = 12\Omega + 6\Omega = \mathbf{18\ \Omega}$</td>
</tr>
</table>
</td>
</tr>
<tr>
<td>15</td><td>A</td><td>64</td>
<td>Voltage dissipated in lamp
V = ?   I = 0.5 A   R = 8.0 Ω
V = I R
V = 0.5 x 8.0
V = 4.0 V

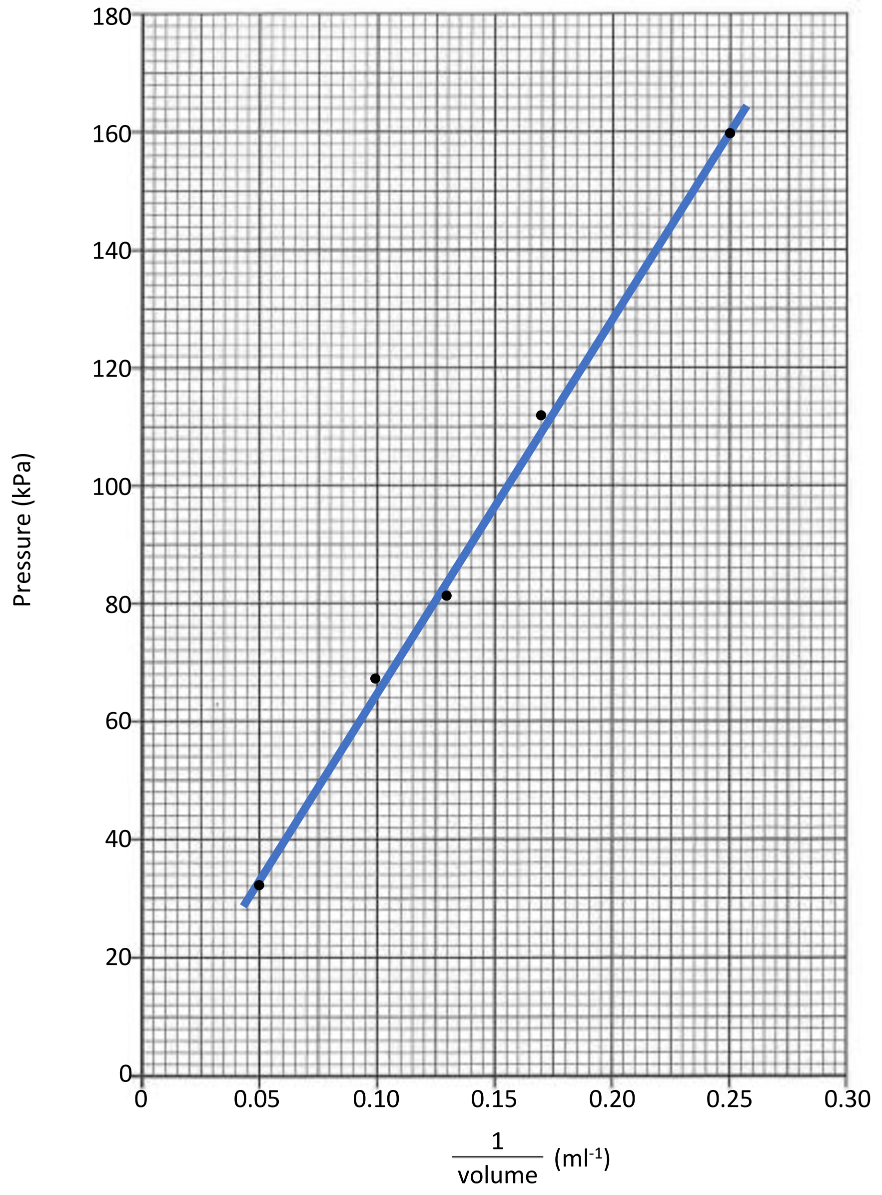
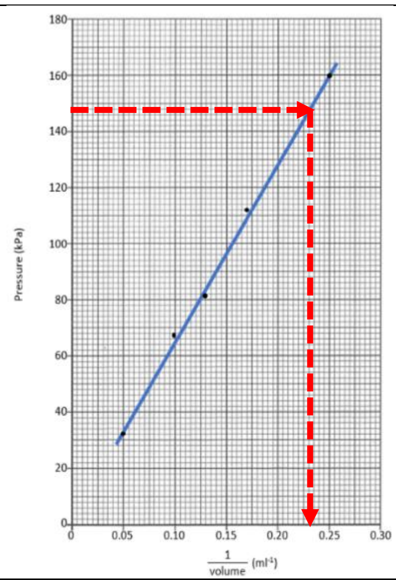
P = ?   V = 4.0 V   R = 8.0 Ω
$P = \frac{V^2}{R} = \frac{(4)^2}{8.0} = \frac{16}{8.0} = 2.0\ W$</td>
</tr>
<tr>
<td>16</td><td>B</td><td>65</td>
<td>
<table>
<tr><th>Kettle = 2.6kW = 2600W</th><th>Television = 170W</th><th>Microwave = 900W</th></tr>
<tr><td>$I = \frac{P}{V} = \frac{2600}{230} = 11.30A$</td><td>$I = \frac{P}{V} = \frac{170}{230} = 0.74A$</td><td>$I = \frac{P}{V} = \frac{900}{230} = 3.91A$</td></tr>
<tr><td>13A fuse required. Power rating is above 720W</td><td>3A fuse required. Power rating is below 720W</td><td>13A fuse required. Power rating is above 720W</td></tr>
</table>
</td>
</tr>
<tr>
<td>17</td><td>C</td><td>34</td>
<td>
<table>
<tr><th>Statement I - Correct</th><th>Statement II - Correct</th><th>Statement III - Incorrect</th></tr>
<tr><td>Insulation of the beaker would reduce the heat loss to the surroundings by trapping a layer of air around the beaker</td><td>Moving the immersion heater further into the water should increase the heat transferred into the water from the heater.</td><td>The joulemeter will directly measure the quantity of energy transferred to the water without the need to calculate E from E = P t</td></tr>
</table>
</td>
</tr>
<tr>
<td>18</td><td>C</td><td>54</td>
<td>
<table>
<tr><th>Statement I - Correct</th><th>Statement II - Incorrect</th><th>Statement III - Correct</th></tr>
<tr><td>The substance evaporates/boils and condenses at 120°C.
The temperature does not change during a change of state.</td><td>The conversion of gas to liquid is complete at 40 minutes so it will be 100% liquid at 40 minutes</td><td>The substance freezes and melts at 120°C.
The temperature does not change during a change of state.</td></tr>
</table>

</td>
</tr>
<tr>
<td>19</td><td>D</td><td>31</td>
<td>P = ?   F = 520N   A = 1.6x10^{-3} m^2 per shoe ∴ 3.2x10^{-3} m^2
$P = \frac{F}{A} = \frac{520}{3.2\times10^{-3}} = 1.6\times10^{5}\ Pa$</td>
</tr>
</table>

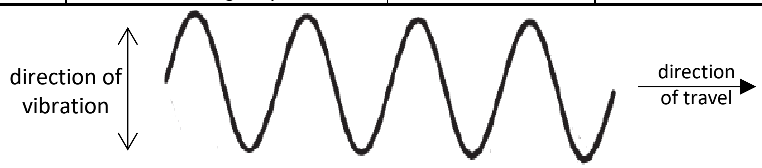
<table>
<tr><td>20</td><td>D</td><td>41</td><td>Volume: Constant $p_1 = 1.90\times10^5$ Pa $T_1 = 20^oC = 293$ K
p_2 = ? $T_2 = 35^oC = 308$ K
$$\frac{p_1}{p_2} = \frac{T_1}{T_2}$$
$$\frac{1.90\times10^5}{p_2} = \frac{293}{308}$$
$$p_2 \times 293 = 1.90\times10^5 \times 308$$
$$p_2 = \frac{1.90\times10^5 \times 308}{293}$$
$$p_2 = 2.0\times10^5 \text{ Pa}$$</td></tr>
<tr><td>21</td><td>B</td><td>83</td><td><table><tr><th>Wavelength</th><th>Amplitude</th></tr><tr><td>3 waves = 120mm
1 wave = $^{120}/_{3}$ = 40mm</td><td>Amplitude $= \frac{0.2}{2} =$ 0.1m</td></tr></table></td></tr>
<tr><td>22</td><td>B</td><td>56</td><td>☒A gamma rays are shorter than visible light and diffract less than visible light
☑B infrared diffracts more than visible as it is longer and has greater frequency
☒C ultraviolet rays are shorter than visible light and diffract less than visible light
☒D radio waves have a lower frequency than microwave radiation
☒E X-rays rays are shorter than visible light and diffract less than visible light
<table><tr><th>EM Type</th><th>Gamma</th><th>X-Ray</th><th>Ultra-violet</th><th>Visible</th><th>Infra-Red</th><th>Microwave</th><th>Radio & TV</th></tr><tr><td>Energy</td><td>High</td><td colspan="5">←→</td><td>Low</td></tr><tr><td>Frequency</td><td>High</td><td colspan="5">←→</td><td>Low</td></tr><tr><td>Wavelength</td><td>Low</td><td colspan="5">←→</td><td>High</td></tr></table><table><tr><td></td><td></td><td rowspan="2">Wavelength must be same before and after obstacle.
Longer waves diffract more than shorter waves</td></tr><tr><td>Long Wave diffraction</td><td>Short Wave diffraction</td></tr></table></td></tr>
<tr><td>23</td><td>E</td><td>49</td><td>A = 5.2 MBq = 5.2×10^6 Bq N = ? t = 1.2 hours = 1.2 x 60 x 60 s
$$A = \frac{N}{t} \quad \therefore N = A \quad t$$
$$N = 5.2\times10^6 \times 1.2 \times 60 \times 60$$
$$N = 2.2\times10^{10}$$</td></tr>
<tr><td>24</td><td>A</td><td>69</td><td>$\dot{H} = 6.0\ \mu Sv\ h^{-1}$ H = 480 μSv t = 5 minutes = 5x60 s
$$\dot{H} = \frac{H}{t}$$ (1 mark)
$$\dot{H} = \frac{480}{5\times60}$$ (1 mark)
$$\dot{H} = 1.6\ \mu Sv\ s^{-1}$$ (1 mark)</td></tr>
<tr><td>25</td><td>B</td><td>87</td><td>☒A Fission: large nucleus splits into two or more smaller nuclei
☑B Fusion: two nuclei of smaller mass combine to produce a nucleus of larger mass
☒C Alpha decay: helium nucleus emitted from a nucleus
☒D Beta Decay: fast electron emitted from a nucleus
☒E Gamma Decay: electromagnetic radiation emitted from a nucleus</td></tr>
</table>

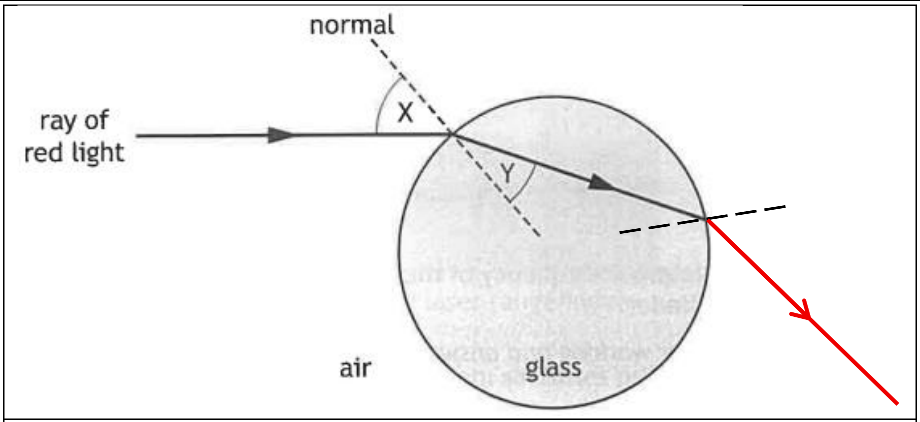
Question	Answer	Physics Covered
1a(i)	39m	Horizontal displacement = 15m (WEST or 270) Vertical displacement = 32m + 4m = 36m (North or 000)  $x = \sqrt{(36)^2 + (15)^2}$ $x = \sqrt{1296 + 225}$ $x = \sqrt{1521}$ $x =$ 39m
1a(ii)	337	$\tan\theta = \frac{opp}{adj} = \frac{15}{36} = 0.417 \quad \therefore \quad \theta = 22.6^{o}$ Bearing = $360^{o} - 23^{o} = 337^{o}$
1b	0.71 m s^{-1} at 337	d = 39 m $\bar{v}$ = ? t = 55 s d = $\bar{v}$ t (1 mark) 39 = $\bar{v}$ x 55 (1 mark) $\bar{v}$ = $^{39}/_{55}$ = 0.71 m s^{-1} (1 mark)
1c	3500 J	E_w = ? F = 68N d = 32+15+4 = E_w = F d (1 mark) E_w = 68 x 51 (1 mark) E_w = 3468 N (1 mark)
2a(i)	$1.40\text{x}10^{7}$ N	W = ? m = $1.43\text{x}10^{6}$ kg g = 9.8 N kg^{-1} W = m x g (1 mark) W = $1.43\text{x}10^{6}$ x 9.8 (1 mark) W = $1.40\text{x}10^{7}$ N (1 mark)
2a(ii)	6.15 m s^{-2}	a = ? F = $2.28\text{x}10^{7} - 1.40\text{x}10^{7} = 0.88\text{X}10^{7}$ N m = $1.43\text{x}10^{6}$ kg $a = \frac{F}{m} = \frac{0.88\text{x}10^{7}}{1.43\text{x}10^{6}} = 6.15$ m s^{-2} (1 mark) (1 mark) (1 mark)
2b(i)	Diagram showing:	<u>Direction Up</u> ↑ One from: Friction Air resistance (1 mark)  (1 mark) <u>Direction Down</u> ↓ One from: weight pull of gravity gravitational pull force due to gravity'
2b(ii)	Answer to Include:	1st mark: air resistance/friction/upwards force/drag *increases* 2nd mark: produces a new unbalanced force upwards
2b(iii)	Answer to Include:	1st mark: upward force = 9300N 2nd mark: Constant velocity (means that forces must be balanced/equal but oppsitie.

3	Open ended question: 0 marks 16% | 1 marks 48% | 2 marks 27% | 3 marks 9%	**1 mark**: Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem. **2 marks**: Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem. **3 marks**: Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.
4a	Answer to include:	1st mark: It experiences a (gravitational) slingshot/catapult (from Mars) 2nd mark: (causing) an increase in its speed/velocity/kinetic energy.
4b(i)	Answer to include:	Small unbalanced force will produce a small acceleration which will continuously increase the velocity over extended/long period of time.
4b(ii)	Answer to include:	The further the distance from the sun the less power produced by solar cells.
4c(i)	1.4 days	41 orbits $\longleftrightarrow$ 56 days 1 orbit $\longleftrightarrow$ 56 days $\times \frac{1}{41}$ = 1.37 days
4c(ii)	Orbital period decreases close to Psyche	The closer an object is to the object it is orbital, the lower the period is as it has to travel a higher velocity to stay in that orbit.
5a	The positive charges in metal dish **or** has same charge as dome **or** like/same charges repel	As electrons are withdrawn from the drawn to create the positive charge in the dome, the foil dishes becoming increasing positively charged. The dishes repel each other and one by one the dishes are pushed upwards and fall.
5b(i)	3.1×10^{-3} A	$Q = 2.50\times10^{-6}$ C $I = ?$ $t = 0.80$ ms $= 0.8\times10^{-3}$ s $Q = I t$ (1 mark) $2.50\times10^{-6} = I \times 0.8\times10^{-3}$ (1 mark) $I = \frac{2.50\times10^{-6}}{0.8\times10^{-3}} = 3.13\times10^{-3}$ A (1 mark)
5b(ii)	1.56×10^{13}	number of electrons $= \frac{\text{discharge}}{\text{charge on 1 electron}} = \frac{2.50\times10^{-6}}{1.60\times10^{-19}} = 1.56\times10^{13}$ electrons
6a	Diagram showing:	 1 mark: Ammeter connected in series. Ammeter can be moved to any point in circuit diagram represented by green connecting leads. 1 mark: Voltmeter must be connected in parallel around the resistor R. No other possible position for voltmeter. Connecting leads must complete the circuit to be correct
6b	56.7 Ω Range Accepted 54.5-57.5	 From graph: When current = 0.02 A then Voltage = 1.3V When current = 0.05 A then Voltage = 3.0V $R = \frac{y_2 - y_1}{x_2 - x_1}$ $R = \frac{3.0 - 1.3}{0.05 - 0.02}$ $R = \frac{1.7}{0.03}$ $R = 56.7\ \Omega$

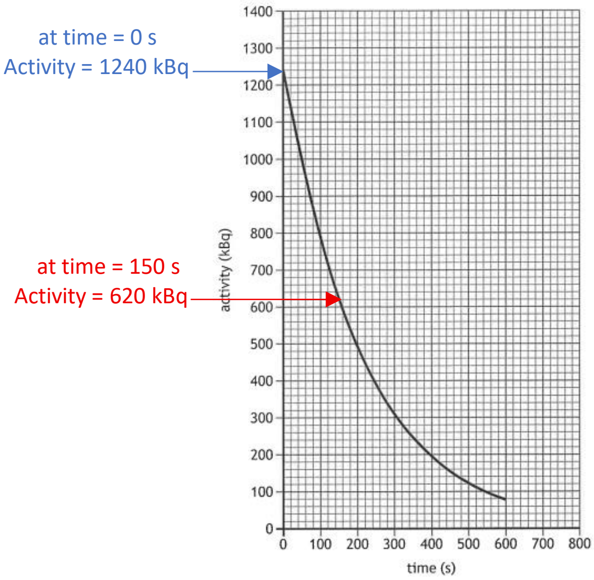
6c	Resistance of lamp is changing/increasing	As the temperature of the lamp in increasing, the resistance of the filament increases. The resistance of the lamp must be increasing as the gradient of the line on the is increasing. The gradient of the line is the resistance due to the equation $R = {}^{V}/_{I}$.
7a	One spotlight goes out rest stay on	The LED spotlights are arranged in a parallel circuit so that if one spotlight is broken then the remaining spotlight stay working as it is connected in parallel.
7b	1.6 A	P = 4.8 x 4 = 19.2W V = 12 V I = ? $P = I \quad V$ (1 mark) $19.2 = I \times 12$ (1 mark) $I = \frac{19.2}{12} = 1.6\ A$ (1 mark)
7c(i)	Mosfet or transistor	
7c(ii)	Answer to include:	1st mark: (when light decreases) resistance in LDR increases 2nd mark: When voltage across the LDR increases 3rd mark: (when voltage across LDR reaches a certain value) Mosfet switches on
8	Open ended question: 0 marks: 39%; 1 marks: 37%; 2 marks: 18%; 3 marks: 7%	**1 mark:** Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem. **2 marks:** Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem. **3 marks:** Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.
9a(i)	35000 J	E_h = ? $c = 4180\ J\ kg^{-1}\ {}^{o}C^{-1}$ m = 0.38 kg $\Delta T = 22^{o}C$ $E = c \times m \times \Delta T$ (1 mark) $E = 4180 \times 0.38 \times 22$ (1 mark) E = 34945 J (1 mark)
9a(ii)	290 s	P = 120 W E = 35000 J t = ? $P = \frac{E}{t}$ (1 mark) $120 = \frac{35000}{t}$ (1 mark) $t = \frac{35000}{120} = 290\ s$ (1 mark)
9a(iii)	One example from:	Insulate the machine better | Put a lid on
9b	0.0458kg	E = 15.3kJ = 15300J m = ? $l = 3.34 \times 10^{-5}\ J\ kg^{-1}$ $E = m \times l$ (1 mark) $15300 = m \times 3.34 \times 10^{-5}$ (1 mark) $m = \frac{15300}{3.34 \times 10^{-5}} = 0.0458\ kg$ (1 mark)
10a	Particles hit the walls of the syringe	Pressure is caused by the particles of gas in air hitting the walls of the container. The greater the pressure the greater the number of collisions.

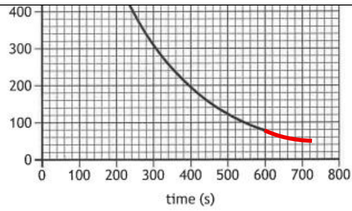
10b(i)	Line graph as shown	 **1 mark** suitable scales, labels and units **1 mark** all points plotted accurately to ± half a division **1 mark** best fit straight line
10b(ii)	Pressure is inversely proportional to volume	The graph shows a directly proportional relationship between $^1/_{volume}$ and the pressure. This means the volume is inversely proportional to the pressure which results in an increase in volume giving a decrease in pressure and vice versa.
10b(iii)	4.3ml	 when pressure = 148 kPa then $^1/_{volume}$ = 0.23 ml^{-1} then volume = $^1/_{0.23ml^{-1}}$ = 4.3ml

10c	One answer from:	Repeat (measurements) and average. | Repeat (measurements) to identify outliers/rogue points. | Increase the range of volumes. | Increase the number of different volumes
11a	Direction of vibration at right angles to the direction of wave travel.	**Transverse Waves** 
11b(i)	Working showing:	N = 500 t = 30minutes = 30x60 s f =? $f = \frac{N}{t} = \frac{500}{30 \times 60} = 0.28\ Hz$ (1 mark) (1 mark)
11b(ii)	$5.0\ m\ s^{-1}$	d = 160 m v = ? t =32 s $d = v \times t$ (1 mark) $160 = v \times 0.32$ (1 mark) $v = {}^{160}/_{0.32} = 5.0\ m\ s^{-1}$ (1 mark)
11b(iii)	18.0 m	$v = 5.0\ m\ s^{-1}$ f = 0.278 Hz λ = ? $v = f \times \lambda$ (1 mark) $5.0 = 0.278 \times \lambda$ (1 mark) $\lambda = {}^{5.0}/_{0.278} = 18.0\ m$ (1 mark)
12a(i)	0.067 s	d = 20200 km = 2.02×10^{7}m $v = 3.0 \times 10^{8}\ m\ s^{-1}$ t =? $d = v \times t$ (1 mark) $2.02 \times 10^{7} = 3.0 \times 10^{8} \times t$ (1 mark) $t = \frac{2.02 \times 10^{7}}{3.0 \times 10^{8}}$ $t = 0.067s$ (1 mark)
12a(ii)	Answer to include:	1 mark: Not geostationary 1 mark: It does not have an (orbital) period of 24 hours | Or | It is not at an (orbital) altitude of 36000km
12b(i)	one answer from:	CCD | Photodiode | phototransistor
12b(ii)	3.3×10^{14} Hz	$v = 3.0 \times 10^{8}\ m\ s^{-1}$ f = ? λ = 904 nm = 904×10^{-9} m $v = f \times \lambda$ (1 mark) $3.0 \times 10^{8} = f \times 904 \times 10^{-9}$ (1 mark) $f = \frac{3.0 \times 10^{8}}{904 \times 10^{-9}}$ $f = 3.32 \times 10^{14}$ Hz (1 mark)
12b(iii)	180 m	d = ? $v = 3.0 \times 10^{8}\ m\ s^{-1}$ t = 1.2×10^{-6} s Total d = $v \times t$ (1 mark) Total d = $3.0 \times 10^{8} \times 1.2 \times 10^{-6}$ (1 mark) Total d = 360 m (1 mark) d = 180m (1 mark)
13a	Answer to include:	X: Angle of incidence | Y: Angle of refraction

<table>
<tr><td>13b</td><td>Answer to include:</td><td><table><tr><td>1 mark</td><td>1 from:<table><tr><td>Refraction takes place</td><td>There is a change in speed.</td><td>The ray is moving from a less (optically) dense into a more (optically) dense medium.</td></tr></table></td></tr><tr><td>1 mark</td><td>The angle of incidence is greater than 0º</td></tr></table></td></tr>
<tr><td>13c</td><td>Diagram showing:</td><td>

Ray emerging from glass changes direction to an angle further from the normal</td></tr>
<tr><td>14a</td><td>Answer to include:</td><td>Alpha and Beta radiation will not travel through the various layers to the surface to be detected</td></tr>
<tr><td>14b(i)</td><td>Time taken for activity to decrease by half</td><td>Half-life is defined as the time taken for the activity/corrected count rate of a radioactive source to half.</td></tr>
<tr><td>14b(ii)</td><td>1 mark:
Sodium-24

1 mark:
Gamma emitter and suitable halflife</td><td><table><tr><th>Source</th><th>Half-Life</th><th>Radiation emitted</th><th>Reasoning</th></tr><tr><td>sodium-24</td><td>15 hours</td><td>beta & gamma</td><td>Source chosen as half-life is not to short or too long and releases gamma radiation which can be detected on the surface</td></tr><tr><td>bismuth-204</td><td>11.2 hours</td><td>beta</td><td>Radiation would be stopped before the surface and not detected from the surface.</td></tr><tr><td>barium-133</td><td>10.5 years</td><td>gamma</td><td>Half-life is too long and would be present in the environment for many years</td></tr><tr><td>barium-137m</td><td>2.6 minutes</td><td>gamma</td><td>Half-life is too short as the radioactivity would decrease too much before all of pipe hade been examined</td></tr></table></td></tr>
<tr><td>14c(i)</td><td>150 s</td><td>
</td></tr>
<tr><td>14c(ii)</td><td>50 kBq

Accept 40-60 kBq</td><td>
</td></tr>
<tr><td>15a(i)</td><td>2.5×10^{-7} Gy</td><td>$H = 0.25\ \mu Sv = 0.25\times10^{-6}$ Sv $D = ?$ $W_r = 1$
$H = D \times W_r$ (1 mark)
$0.25\times10^{-6} = D \times 1$ (1 mark)
$D = 0.25\times10^{-6} = 2.5\times10^{-7}$ Gy (1 mark)</td></tr>
</table>

15a(ii)	1.6×10^{-5} J or 0.000016 J	$D = 2.5 \times 10^{-7}$ Gy $E = ?$ $m = 64$ kg $D = \dfrac{E}{m}$ (1 mark) $\therefore$ $2.5 \times 10^{-7} = \dfrac{E}{64}$ (1 mark) $\therefore$ $E = 2.5 \times 10^{-7} \times 64 = 1.6 \times 10^{-5}$ J (1 mark)
15b	Answer to include:	Process where atoms become charged by losing or gaining electrons
15c	One answer from:	Stand behind screen / Limit time near machine / Keep distance from machine